

UJJAWAL SHRIVASTAV

Full-Stack & Backend Developer — Python · React/Next.js · FastAPI · PostgreSQL

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Portfolio • LinkedIn • GitHub

SUMMARY

Full-Stack Developer (B.Tech CS — 2025) who ships production web apps end-to-end: Next.js + PostgreSQL frontends, FastAPI + asyncio backends, OAuth/OTP auth, Redis rate-limiting, HMAC-verified payments, and 100+ automated tests. Projects span realtime multiplayer, hybrid vector search, and multi-agent REST APIs.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, React 19, Next.js, Node.js, FastAPI, REST API, SSE streaming, asyncio

Backend & Data: PostgreSQL, SQL, pgvector, Drizzle ORM, Pydantic v2, Docker, Git, Vercel, Cloudflare Workers

AI/ML (applied): LLMs (Claude, Gemini), RAG pipelines, scikit-learn, LightGBM, CatBoost, sentence-transformers, embeddings

PROJECTS

EngiNerd — Full-Stack AI Learning Platform

TypeScript, Next.js, Node.js, PostgreSQL, Drizzle ORM, Redis, Playwright • 2026

- Built a full-stack Next.js + PostgreSQL platform with OTP + OAuth auth, HMAC-verified idempotent payments, and tiered Redis rate-limiting — production-hardened across 68 unit and 43 Playwright end-to-end tests.
- Engineered a 5-stage LLM content pipeline (research → mapping → deep-dive → writing → review) with SSE streaming to a live dashboard and an offline stub mode for deterministic CI runs.

McpIndex — Hybrid Search Platform

Next.js, PostgreSQL, pgvector, Gemini Flash, Vercel • 2026

- Shipped a full-stack search platform indexing 1,532 entries with hybrid BM25 + pgvector semantic search and an automated 6-dimensional quality-scoring pipeline; delivered end-to-end in 2 days.

Jarvis — FastAPI Multi-Agent Backend

Python, FastAPI, asyncio, REST API, Pydantic v2, Docker, pytest • 2026

- Designed and deployed a FastAPI REST API daemon (/chat, /task) with asyncio-parallel workers, Pydantic v2 schema validation, and SSE streaming — containerised with Docker, run as systemd services, 35+ pytest tests.
- Built a production RAG pipeline: documents chunked, embedded (all-MiniLM-L6-v2), stored in ChromaDB, retrieved by cosine similarity; four-rubric self-correction loop gates every response for quality.

Math Tug — Real-Time 2-Player Web App

React 19, TypeScript, Tailwind 4, PartyKit (Cloudflare Workers), Capacitor • 2026

- Built a React 19 + TypeScript multiplayer game with authoritative WebSocket sync on Cloudflare Workers and a strict TypeScript state machine — single codebase ships to web and Android via Capacitor.

EDITH — Full-Stack Agentic Copilot (Tata Steel Hackathon, Rank 22 → Round 2)

Python, FastAPI, Next.js, React 19, LightGBM, CatBoost, SSE • 2026

- Built a FastAPI + Next.js full-stack copilot with a five-agent reasoning chain, SSE-streamed live asset monitoring, and a LightGBM + CatBoost ML ensemble — ranked 22nd nationally across 70+ submissions.

EDUCATION

B.Tech — Computer Science (AI/ML Specialization), Krishna Engineering College

2022 – 2025

STRENGTHS & INTERESTS

Ships full-stack features end-to-end · strong on backend reliability (auth, rate-limiting, testing) · learns new stacks fast |
Interests: open-source tools · side projects · snooker